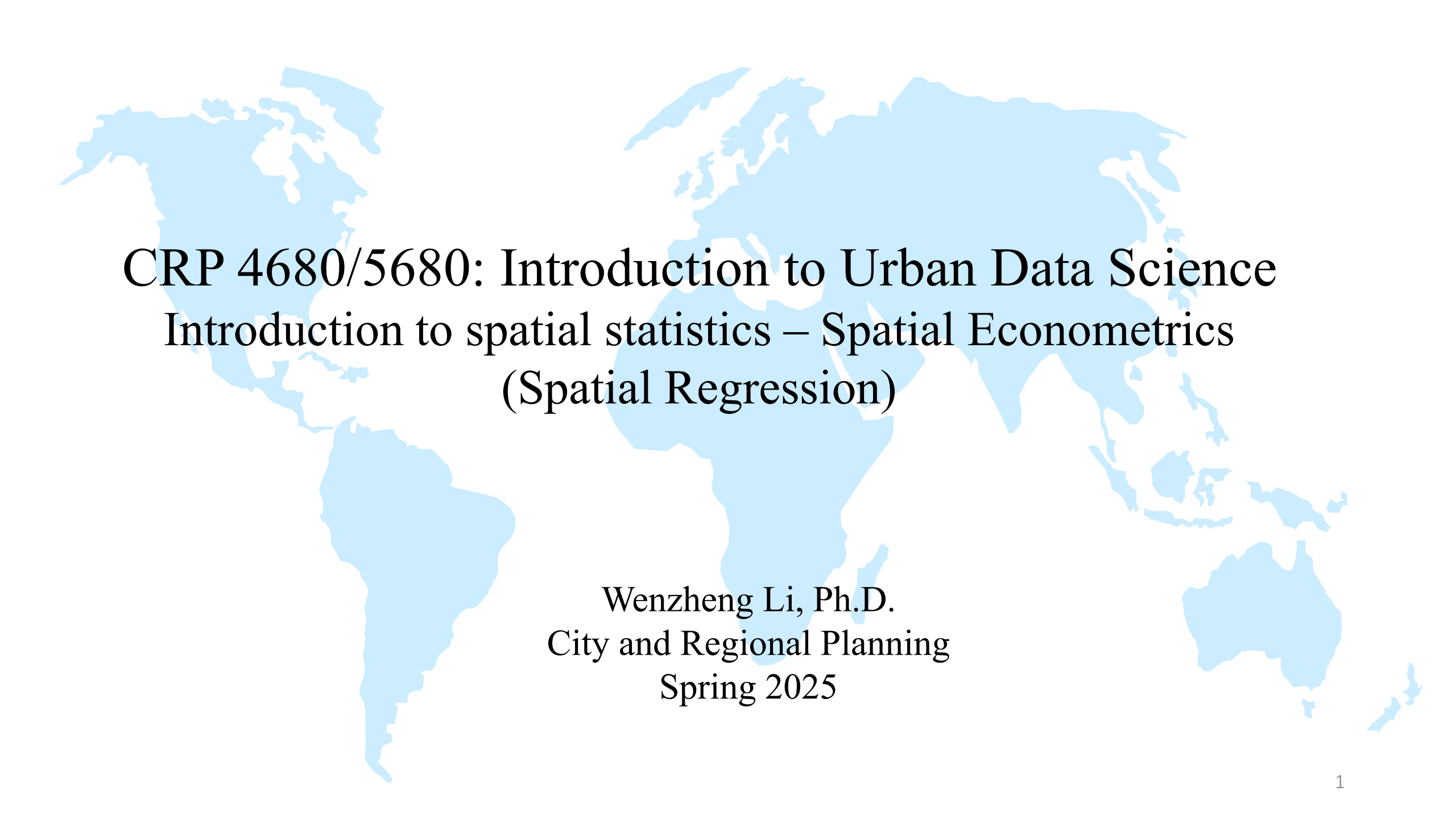




CRP 4680/5680: Introduction to Urban Data Science

Introduction to spatial statistics – Spatial Econometrics (Spatial Regression)

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City and Regional Planning
Spring 2025

OUTLINE – Spatial Econometrics

Recap: Ordinary Least Square Regression

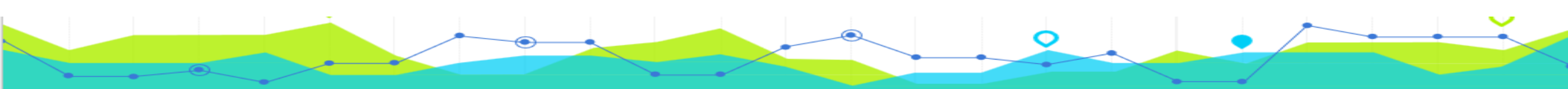
What is spatial regression? How does it matter?

Spatial regression models that capture spatial dependence

- spatial lag model (SLM)
- spatial error model (SEM)
- spatially lagged X model (SLX)

After spring Break: Spatial regression models that capture spatial heterogeneity

- Geographically weighted regression (GWR): a spatial regression technique that allows model parameters to vary across geographic space





Recap: OLS regression

1

From Economic Model

- Test economic theory

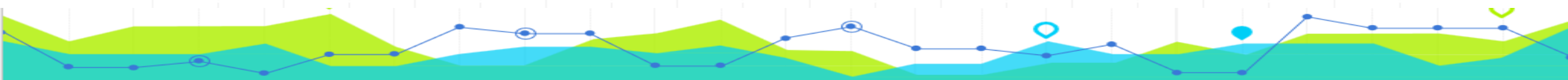
$$y = f(x_1, x_2, x_3, x_4, \dots, x_n)$$


What effect x_1 has on y

- Any effect? (How different from "0")
- How large? (Evaluate the magnitude)

- ▶ Example: Hedonic theory on housing price

$$price = f(structural, locational, neighborhood)$$



To Econometric model

► Economic model

Specify

$f()$

► Econometric model

$$y = f(x_1, x_2, x_3, x_4, x_5)$$

Measures of the theoretical variables
(Observed)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + u$$

Parameters: Describe the direction and strengths
of the relationship between y and x

Error term (disturbance term):

- Contains unobserved factors
- Can never eliminate u entirely

Simple Regression Model

$$y = \beta_0 + \beta_1 x_1 + u$$

y

Dependent variable
Explained variable
Response variable
Predicted variable
Regressand

x_1

Independent variable
Explanatory variable
Control variable
Predictor variable
Regressor



What do we care about?

$$y = f(x_1, x_2, \dots, x_n)$$

Parameterization

Linear assumption

$$y = \beta_0 + \beta_1 \times x_1 + u$$

$$\text{Housing Price} = \beta_0 + \beta_1 \times \text{Subway} + u$$

59,000
23,000
40,000
...

True value

547
2050
730
...

Residuals:
observed y –
predicted y

Estimators

$$\widehat{\text{Housing Price}} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{Subway}$$

$$\widehat{\text{Housing Price}} = 44850 + 5.0906 \times \text{SDIST}$$

Estimates

Unbiased:

$$E(\hat{\beta}_0) = \beta_0$$

$$E(\hat{\beta}_1) = \beta_1$$

- This means that, on average, the estimator will correctly estimate the true parameter value.

Consistent: as the sample size gets larger

$$\hat{\beta}_0 \rightarrow \beta_0$$

$$\hat{\beta}_1 \rightarrow \beta_1$$

- as the sample size (n) increases, the estimator converges to the true parameter value.

Efficient: smaller variance and standard errors

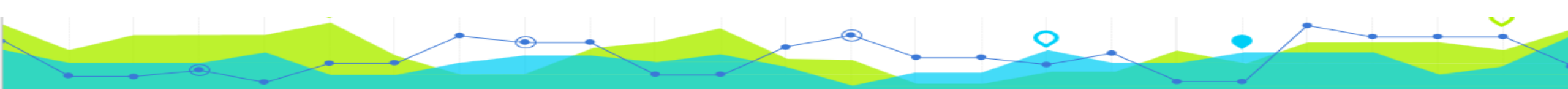
$$\text{Var}(\hat{\beta}_0)$$

$$\text{Var}(\hat{\beta}_1)$$

- A more efficient estimator means smaller standard errors, which leads to narrower confidence intervals and more precise inferences.

Linear Regression Assumptions (cont.)

- **Linearity** - the relationships between the predictors and the outcome variable should be linear
- **Low multicollinearity** - we avoid predictors that are highly collinear, which can cause problems when estimating regression coefficients.
- **Independence** - the errors associated with one observation are not correlated with the errors of any other observation. If errors are correlated, it suggests issues such as autocorrelation (in time series) or spatial dependence (in spatial data).
- **Constant error variance (homoscedasticity)**; the error variance should be constant. The opposite is called heteroscedasticity.
- **Model misspecification**: The model is specified properly – **all known variables that are relevant predictors of the dependent variable are included. This means no omitted variable bias.** There are not individual observations that exert undue influence on the coefficients.



Linear Regression Assumptions

- **Linearity** - the relationships between the predictors and the outcome variable should be linear

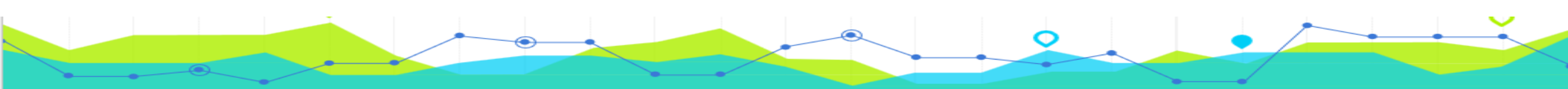
Which one violates the linearity assumption?

Linearity assumption means the model must be **linear in parameters** (i.e., the β s), not necessarily in the variables.

$$y = \beta_0 + \beta_1 X + \beta_2 X_2 + \epsilon$$

$$y = \beta_0 + \beta_1 X_2 + \beta_2 X_2^2 + \epsilon$$

$$y = \beta_0 + \beta_1 X_2 + \beta_2^2 X_2 + \epsilon$$



Linear Regression Assumptions

- **Low multicollinearity** - Multicollinearity occurs when two or more predictor variables (independent variables) are highly correlated with each other.
- When multicollinearity is high, it becomes difficult to determine the individual effect of each predictor on the dependent variable because they share similar information

How to address?

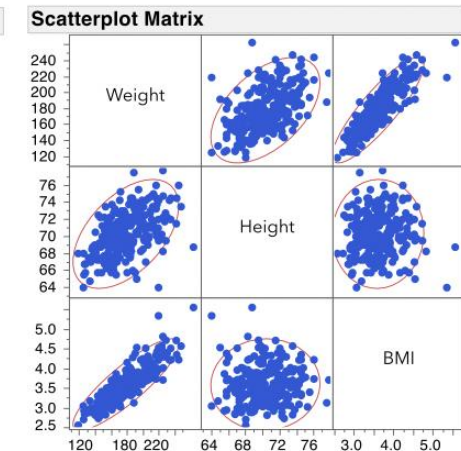
- I will first create a scatterplot matrix to observe the relationships between variables, then perform a Variance Inflation Factor (VIF) test. A **rule of thumb** is that **VIF > 5** indicates problematic multicollinearity.

In this week's Lab, we learn:

- The Multicollinearity Condition Number (MCN)

MCN Value	Multicollinearity Level	Interpretation
< 10	Low	No concern — predictors are not highly correlated
10–30	Moderate	Some correlation between predictors — may cause instability
> 30	High / Severe	Strong multicollinearity — coefficients may be unreliable

Multivariate			
Correlations			
Weight	1.0000	0.5129	0.8668
Height	0.5129	1.0000	0.0220
BMI	0.8668	0.0220	1.0000



Linear Regression Assumptions

- **Constant error variance (homoscedasticity):** the error variance should be constant. The opposite is called heteroscedasticity. Mathematically, $Var(\varepsilon|X) = \sigma^2$.

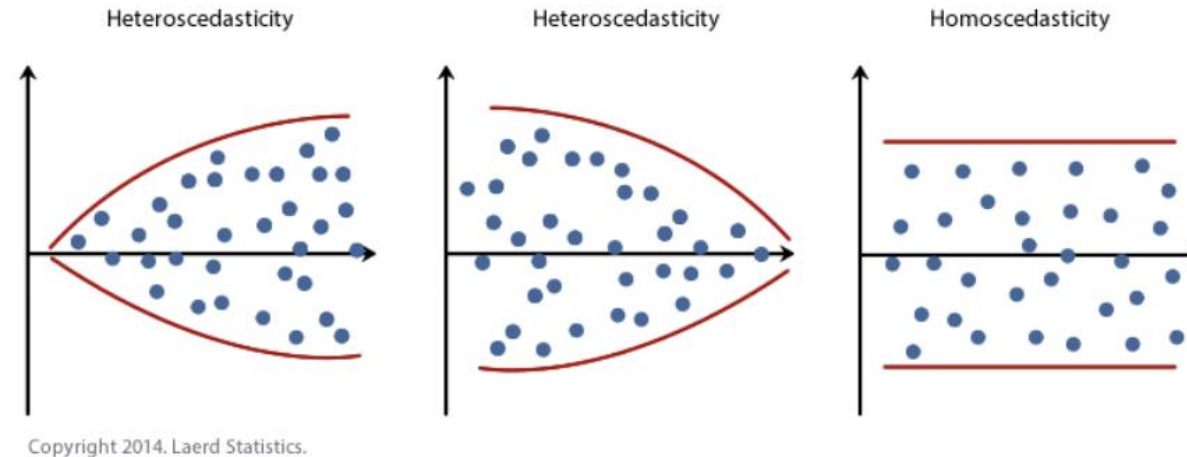


Image Source: Laerd Statistics

The above diagram shows the difference between Homoscedasticity and Heteroscedasticity. The variance of errors is constant in case of homoscedasticity while it's not the case if errors are heteroscedastic.

Common Solutions

- Robust Standard Errors:
 - It doesn't change the coefficients from OLS but adjusts the standard errors to account for heteroskedasticity.
- Add Missing Explanatory Variables
 - check omitted variable bias
- Generalized Least Squares (GLS)
 - If you can correctly specify the variance structure, GLS gives **efficient** estimates.
- Transform the Dependent Variable:
 - taking the **log**

Linear Regression Assumptions

Independence - No Autocorrelation (i.e. independence of Errors)

- Check for spatial autocorrelation: Use Moran's I
- Apply Spatial Regression Models:
 - Spatial Lag Model (SLM): Accounts for spatial spillover effects by including the spatially lagged dependent variable.
 - Spatial Error Model (SEM): Corrects for spatially autocorrelated error terms.
- Cluster-Robust Standard Errors: Adjust standard errors to account for spatial clustering.



Model misspecification – model choice

- What variables should be included (theoretically)?
- The influential factors of housing price? (hedonic price model)
 - Locational attributes
 - Structural attributes
 - Neighborhood attributes



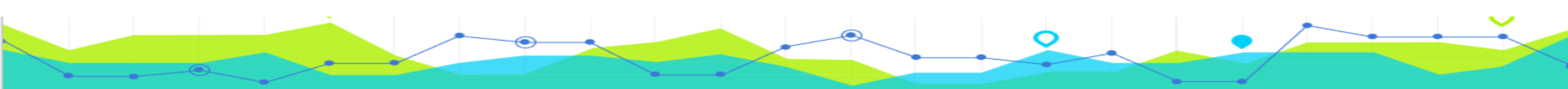
TABLE 1: List of Commonly Used Housing Attributes in Hedonic Price Models.

Attribute		Expected effect on housing price
Locational	Distance from CBD	-ve
	View of the sea, lakes or rivers	+ve
	View of hills/valley/golf course	+ve
	Obstructed view	-ve
	Length of land lease	+ve
Structural	Number of rooms, bedrooms, bathrooms	+ve
	Floor area	+ve
	Basement, garage, and patio	+ve
	Building services (e.g. lift, air conditional system etc)	+ve
	Floor level (multi-storey buildings only)	+ve
	Structural quality (e.g., design, materials, fixtures)	+ve
	Facilities (e.g., swimming pool, gymnasium, tennis court)	+ve
	Age of the building	-ve
Neighbourhood	Income of residents	+ve
	Proximity to good schools	+ve
	Proximity to Hospitals	?
	Proximity to Places of worship (e.g., mosques, churches, temples)	+ve
	Crime rate	-ve
	Traffic/airport noise	-ve
	Proximity to Shopping centers	?
	Proximity to Forest	?
	Environmental quality (e.g., landscape, garden, playground)	+ve

+ve – positive impact on housing prices; -ve – negative impact on housing prices

? – varies from place to place, the actual effect is an empirical question

Source: Chau, K. W., & Chin, T. L. (2002). A critical review of literature on the hedonic price model.



Misspecification of Variables

Situation	Misspecification	Potential Consequences for Key Variables' Estimates
An irrelevant independent variable is included	Over-specification	Inefficient Unbiased and consistent
A relevant independent variable is omitted	Under-specification	Biased and inconsistent

Misspecification of Variables

$$\widehat{\text{Housing Price}} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{Subway}$$

$$\widehat{\text{Housing Price}} = \tilde{\beta}_0 + \tilde{\beta}_1 \times \text{Subway} + \tilde{\beta}_2 \times \text{Bedroom}$$

$$\hat{\beta}_1 = \tilde{\beta}_1 + \tilde{\beta}_2 \hat{\delta}_1$$

the slope from the simple regression of
Bedroom on Subway

$$\widehat{\text{Bedroom}} = \delta_0 + \hat{\delta}_1 \times \text{Subway}$$



	DTUP I	DTUP II	DTUP III	DTUP IIII
CONST	45094.00*** (682.76)	47307.95*** (1703.53)	48953.08*** (1887.08)	56537.19*** (1690.91)
SDIST	-5.51*** (0.46)	-5.29*** (0.46)	-5.53*** (0.48)	-2.19*** (0.45)
FAREA		-93.86*** (26.65)	-97.79*** (26.68)	-55.75** (23.14)
BED		632.40 (935.00)	706.68 (934.30)	-332.66 (809.89)
BATH		4303.29** (1715.49)	4337.44** (1712.95)	3895.09*** (1473.98)
LIV		-762.04 (1139.45)	-716.95 (1137.93)	435.82 (981.06)
BN500			-58.07** (28.85)	-42.01* (24.91)
DIST_CBD				0.57*** (0.11)
DIST_CENTER				-1.61*** (0.12)
N	1000	1000	1000	1000
R2_Adj	0.12	0.14	0.14	0.36

Standard errors in parentheses.
 * p<.1, ** p<.05, ***p<.01

- The estimated coefficient of *SDIST* is stable across the first three models.
- After adding *DIST_CBD* and *DIST_CENTER*, the magnitude of the coefficient of *SDIST* drops.



Basic Matrix Notation

2

Basic Matrix Notation

Matrices are key concepts in mathematics. A **matrix** is a rectangular array of numbers, symbols, or expressions arranged in rows and columns, where each element is identified by its row and column position.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Row (m) →

↓ Columns (n)

Example: $\begin{bmatrix} 6 & 9 \\ 5 & -4 \end{bmatrix}_{2 \times 2}, \begin{bmatrix} 3 & -4 & 5 \\ 1 & 7 & 6 \\ 6 & -2 & 9 \end{bmatrix}_{3 \times 3}$

Basic Matrix Notation

Matrix Addition

$$\begin{matrix} A_{2 \times 2} & & B_{2 \times 2} & & C_{2 \times 2} = A_{2 \times 2} + B_{2 \times 2} \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} & + & \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} & = & \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} \end{matrix}$$

Matrix addition requires matrices of the same dimensions. Each element in the resulting matrix is obtained by adding the corresponding elements from the two matrices.

Matrix subtraction

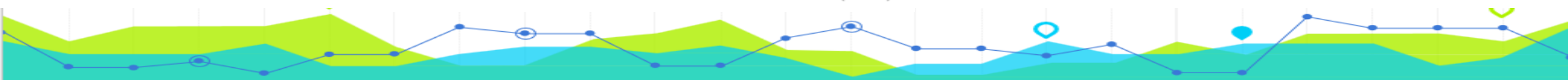
$$\begin{matrix} A_{2 \times 2} & & B_{2 \times 2} & & C_{2 \times 2} = A_{2 \times 2} - B_{2 \times 2} \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} & - & \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} & = & \begin{bmatrix} 1-5 & 2-6 \\ 3-7 & 4-8 \end{bmatrix} \end{matrix}$$

Matrix subtraction follows the same rule as addition but subtracts corresponding elements from the two matrices.

Scalar Multiplication

$$\begin{matrix} A_{2 \times 2} \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \end{matrix} \quad 3 \times A = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 3 & 3 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}_{(2 \times 2)}$$

In scalar multiplication, each element in the matrix is multiplied by the scalar value. This operation maintains the matrix's dimensions.



Basic Matrix Notation - Multiplication

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$$

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} (ap + br) & (aq + bs) \\ (cp + dr) & (cq + ds) \end{bmatrix}$$

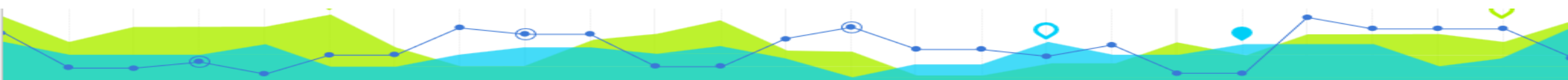
Example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} (1 \times 7 + 2 \times 9 + 3 \times 11) & (1 \times 8 + 2 \times 10 + 3 \times 12) \\ (4 \times 7 + 5 \times 9 + 6 \times 11) & (4 \times 8 + 5 \times 10 + 6 \times 12) \end{bmatrix}$$

$A_{2 \times 3} \qquad B_{3 \times 2} \qquad C_{2 \times 2}$

If Matrix A has dimensions $m \times n$ and Matrix B has dimensions $n \times p$, then their product $C = A \times B$ will have dimensions: $C_{(m \times p)}$

- $A_{m \times n} \times B_{n \times p} = AB_{m \times p}$
- In other words, The number of columns in the first matrix (A: n) must match the number of rows in the second matrix (B: n).



Matrix Notation for OLS regression

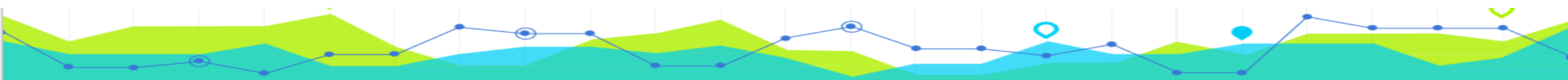
- ▶ When there are only **1** independent variable and a constant

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{21} \\ 1 & x_{31} \\ \vdots & \vdots \\ 1 & x_{n1} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

- ▶ Compact Matrix Notation

$$Y = X\beta + \varepsilon$$

	y		x_1
	DTUP	CONSTANT	SDIST
1	40886.220	1	558.33545
2	27734.275	1	603.46985
3	28393.690	1	865.78906
4	33236.010	1	340.77704
5	33183.953	1	2037.08520
...	...	...	...



Notation

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{21} \\ 1 & x_{31} \\ \vdots & \vdots \\ 1 & x_{n1} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$



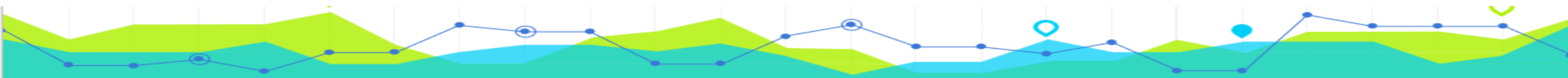
$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \beta_0 + \begin{bmatrix} x_{11} \\ x_{21} \\ x_{31} \\ \vdots \\ x_{n1} \end{bmatrix} \beta_1 + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$



$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \varepsilon_1 \\ \beta_0 + \beta_1 x_{21} + \varepsilon_2 \\ \beta_0 + \beta_1 x_{31} + \varepsilon_3 \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \varepsilon_n \end{bmatrix}$$



$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \beta_0 \\ \beta_0 \\ \beta_0 \\ \vdots \\ \beta_0 \end{bmatrix} + \begin{bmatrix} \beta_1 x_{11} \\ \beta_1 x_{21} \\ \beta_1 x_{31} \\ \vdots \\ \beta_1 x_{n1} \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$



Notation

► Compact Matrix Notation

$$Y = X\beta + \varepsilon$$

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► For each observation i

$$i = 1 \dots n$$

$$y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$$

When there are only
1 independent variable
and a constant



Notation

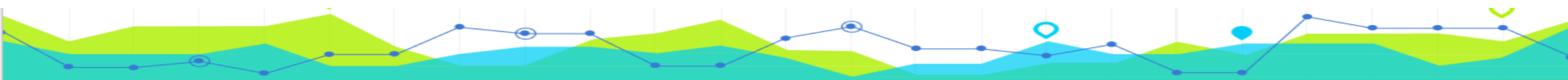
- ▶ When there are **4** independent variables and a constant

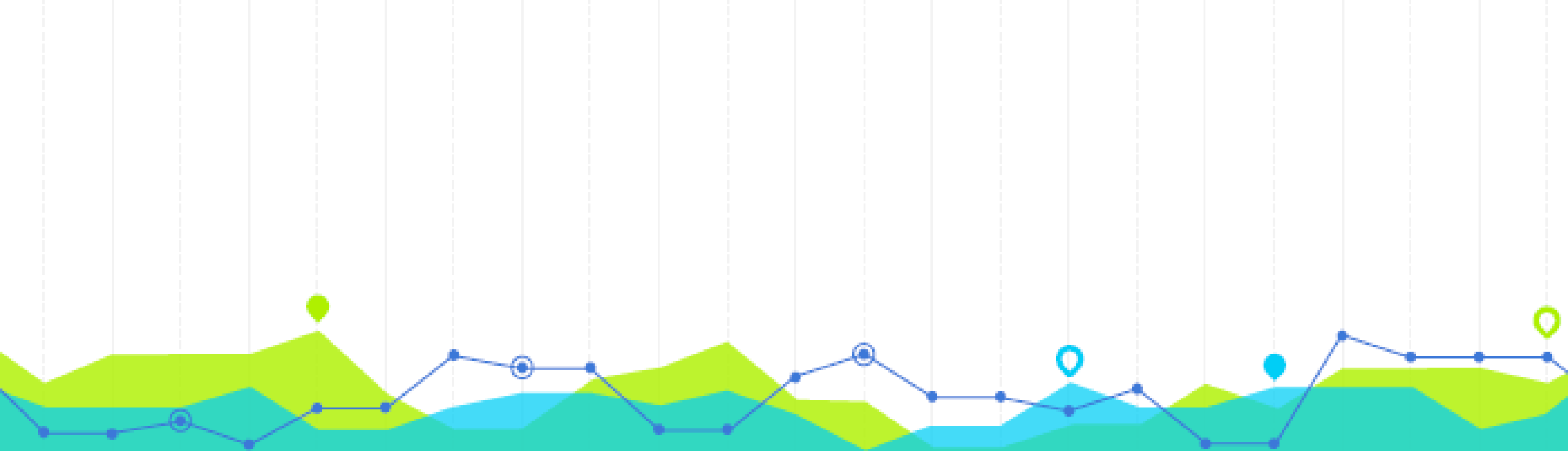
$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & x_{14} \\ 1 & x_{21} & x_{22} & x_{23} & x_{24} \\ 1 & x_{31} & x_{32} & x_{33} & x_{34} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n3} & x_{n4} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

- ▶ Compact Matrix Notation

$$Y = X\beta + \varepsilon$$

	y		x_1	x_2	x_3	x_4
	DTUP	CONSTANT	SDIST	FAREA	BED	BATH
1	40886.220	1	558.33545	131.37	3	2
2	27734.275	1	603.46985	59.59	2	1
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4	33236.010	1	340.77704	135.75	3	2
5	33183.953	1	2037.08520	87.53	2	1
...	...	...	...	...	...	...





Spatial Regression Models (I) 3

Spatial regression: motivations

Question: What determines the price of a single family house?

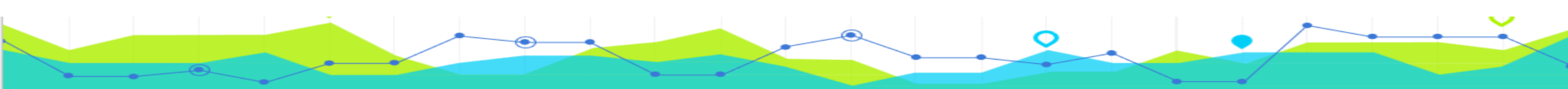
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Spatial regression: motivations

Question: What determines the price of a single family house?

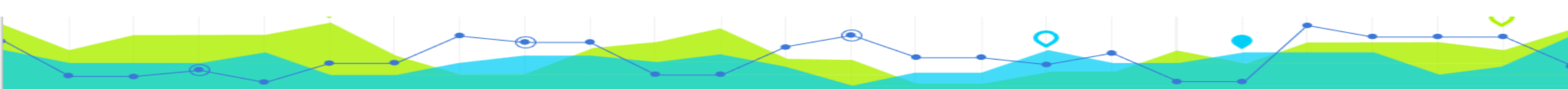
- **Location/Land value + actual house value (e.g., size, age, etc).**
 - Location/Land value:
 - Proximity to: Noise, pollution, other types of land uses
 - Proximity to: Schools/school district, transit, green spaces
 - Price is influenced by these factors through geographic processes.



Spatial regression: motivations

Question: What determines the price of a single family house?

- **Location/Land value + actual house value (e.g., size, age, etc).**
 - Location/Land value:
 - Proximity to: Noise, pollution, other types of land uses
 - Proximity to: Schools/school district, transit, green spaces
 - Price is influenced by these factors through geographic processes.
- **Spatial autocorrelation:** The price of two homes close to each other both influence each other and are influenced by similar explanatory variables. The prices are not independent observations and we need to account for this.



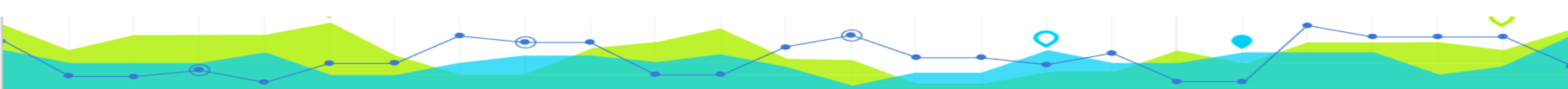
What is spatial regression

- What is spatial regression?
 - spatial regression: applications of statistical methods to understand economic relationship empirically (of which regression is one technique) with explicitly spatial considerations.



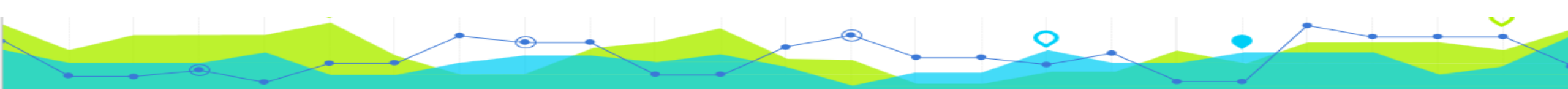
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 - Spatial regression: applications of statistical methods to understand economic relationship empirically (of which regression is one technique) with explicitly spatial considerations.
- The key to spatial regression is incorporating spatial lag terms to capture spatial dependencies and interactions between neighboring observations.
- ***spatial lag*** is essentially the neighborhood influence—the price of a house is influenced by the prices of nearby houses; a region's economic growth, crime rate, or pollution level is often affected by surrounding areas



What is spatial regression

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- ***spatial lag*** is essentially the neighborhood influence—the price of a house is influenced by the prices of nearby houses; a region's economic growth, crime rate, or pollution level is often affected by surrounding areas
- What happens in one place doesn't stay there; it spills over to nearby locations!



Spatial Regression

The key to spatial regression is incorporating spatial lag terms to capture spatial dependencies and interactions between neighboring observations.

▶ Linear OLS Model	$Y = X\beta + \varepsilon$
▶ Spatial Lag Model	$Y = \rho WY + X\beta + \varepsilon$
▶ Spatial Error Model	$Y = X\beta + \lambda W\varepsilon + u$
▶ Spatial Durbin Model	$Y = \rho WY + X\beta + \theta WX + \varepsilon$
▶ Spatial SLX Model	$Y = X\beta + \theta WX + \varepsilon$

For more models, see Figure 2.1 (Page 9) in Elhorst, J. P. (2014). *Spatial econometrics: from cross-sectional data to spatial panels* (Vol. 479, p. 480). Heidelberg: Springer.

Weight Matrix W

3		
1	4	5
2		

$$WY = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 3000 \\ 1000 \\ 1000 \\ 3000 \\ 4000 \end{bmatrix}$$

Neighboring Relationship

1's neighbor
2's neighbor
3's neighbor
4's neighbor
5's neighbor

$$\begin{array}{l} \longrightarrow \\ \longrightarrow \\ \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1000 \\ 2000 \\ 3000 \\ 4000 \\ 5000 \end{bmatrix} =$$

$$\begin{bmatrix} 0 \times 1000 + 1 \times 2000 + 1 \times 3000 + 1 \times 4000 + 0 \times 5000 \\ 1 \times 1000 + 0 \times 2000 + 0 \times 3000 + 0 \times 4000 + 0 \times 5000 \\ 1 \times 1000 + 0 \times 2000 + 0 \times 3000 + 0 \times 4000 + 0 \times 5000 \\ 1 \times 1000 + 0 \times 2000 + 0 \times 3000 + 0 \times 4000 + 1 \times 5000 \\ 0 \times 1000 + 0 \times 2000 + 0 \times 3000 + 1 \times 4000 + 0 \times 5000 \end{bmatrix}$$

Row standardization

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{array}{l} \times \frac{1}{3} \\ \times 1 \\ \times 1 \\ \times \frac{1}{2} \\ \times 1 \end{array}$$

$$= \begin{bmatrix} 2000 + 3000 + 4000 \\ 1000 \\ 1000 \\ 1000 + 5000 \\ 4000 \end{bmatrix} \begin{array}{l} \times \frac{1}{3} \\ \times 1 \\ \times 1 \\ \times \frac{1}{2} \\ \times 1 \end{array}$$

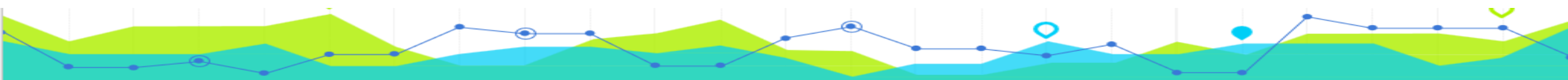
Spatial Lagged Variables

$$WY = \begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 3000 \\ 1000 \\ 1000 \\ 3000 \\ 4000 \end{bmatrix}$$

WX = The spatial lag of explanatory variables, capturing how neighboring values of X influence the dependent variable.

$W\varepsilon$ = The spatial lag of residuals, relevant in spatial error models where error terms are spatially correlated.

- weighted average of neighboring values
- neighbors defined in weight matrix

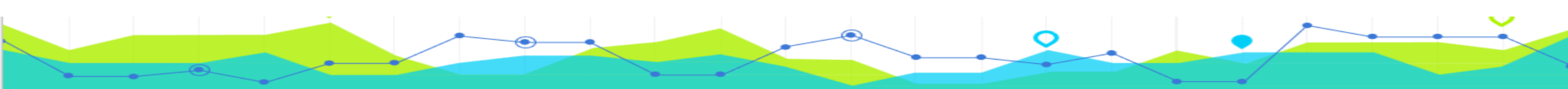


SLX model

$$Y = \alpha + X\beta + WX\gamma + \varepsilon$$

where:

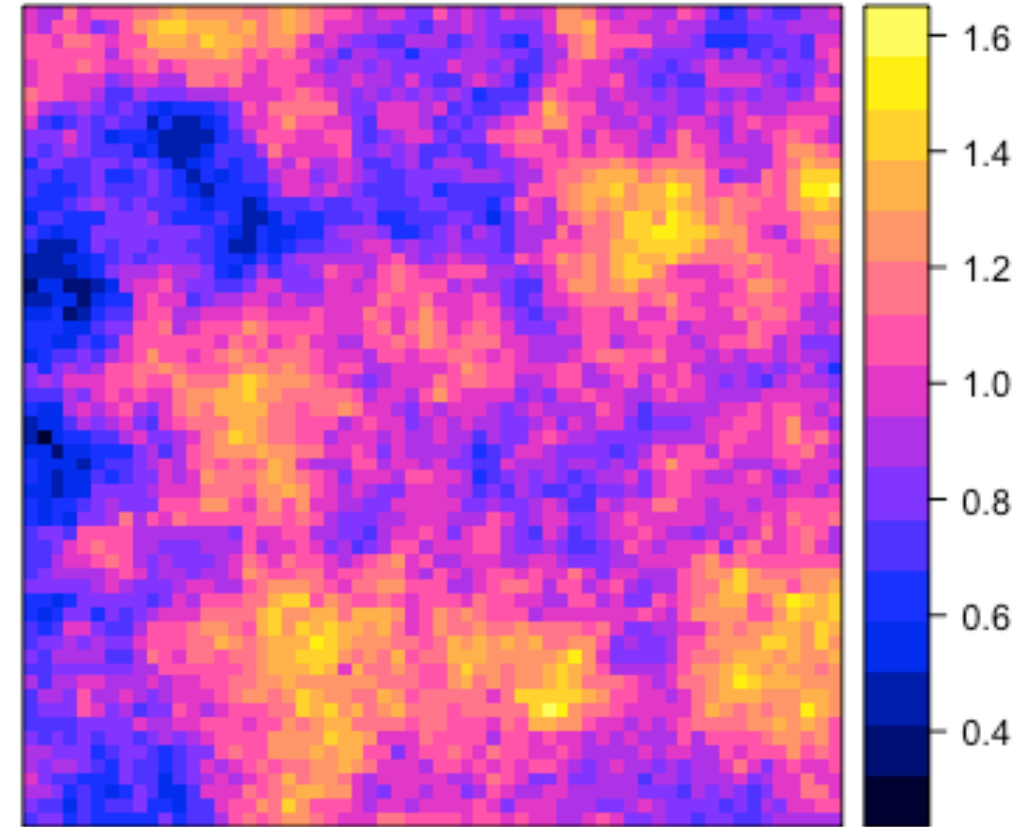
- X is the $n \times k$ matrix of independent variables.
- WX represents the **spatial lag of independent variables**, where W is the $n \times n$ spatial weight matrix that defines spatial relationships.
- β and γ are $k \times 1$ coefficient vectors:
 - β captures the **direct** effect of X on Y .
 - γ captures the **spillover** effect of neighboring values of X on Y .
- ε is the $n \times 1$ vector of error terms.
- **Use When:** You believe that the dependent variable in one location is affected by the values of the neighboring locations in the independent variable.



spatial error model

$$Y = X\beta + \lambda W\varepsilon + u$$

- What if our residuals were spatially autocorrelated?
- We are excluding a process that is producing spatially non-randomness in our outcomes.
- Remember: We want random errors, i.e. "noise" in our residuals
- This actually violates the error term assumptions in a standard regression model, but pysal has some alternate estimations for the same statistics we would normally use.

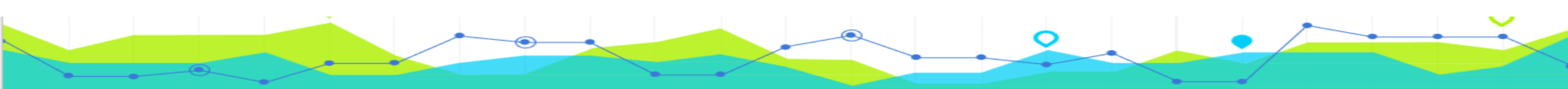


Let's pretend this is a map of residuals: From <https://www.r-bloggers.com/2019/07/exploring-spatial-autocorrelation-in-r/>

spatial lag model

$$Y = \rho WY + X\beta + \varepsilon$$

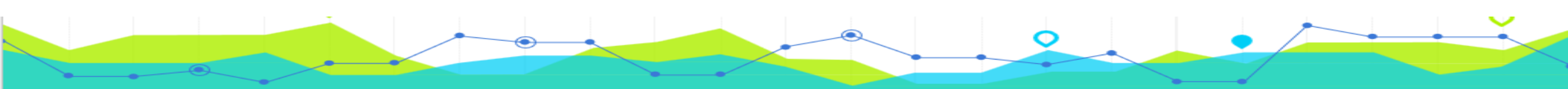
- Y is the $n \times 1$ vector of dependent variable values.
- W is the $n \times n$ spatial weight matrix that defines spatial relationships between observations.
- ρ is the spatial autoregressive parameter, which measures the strength of spatial dependence.
- X is the $n \times k$ matrix of independent variables.
- β is the $k \times 1$ vector of coefficients for the independent variables.
- ε is the $n \times 1$ vector of error terms.

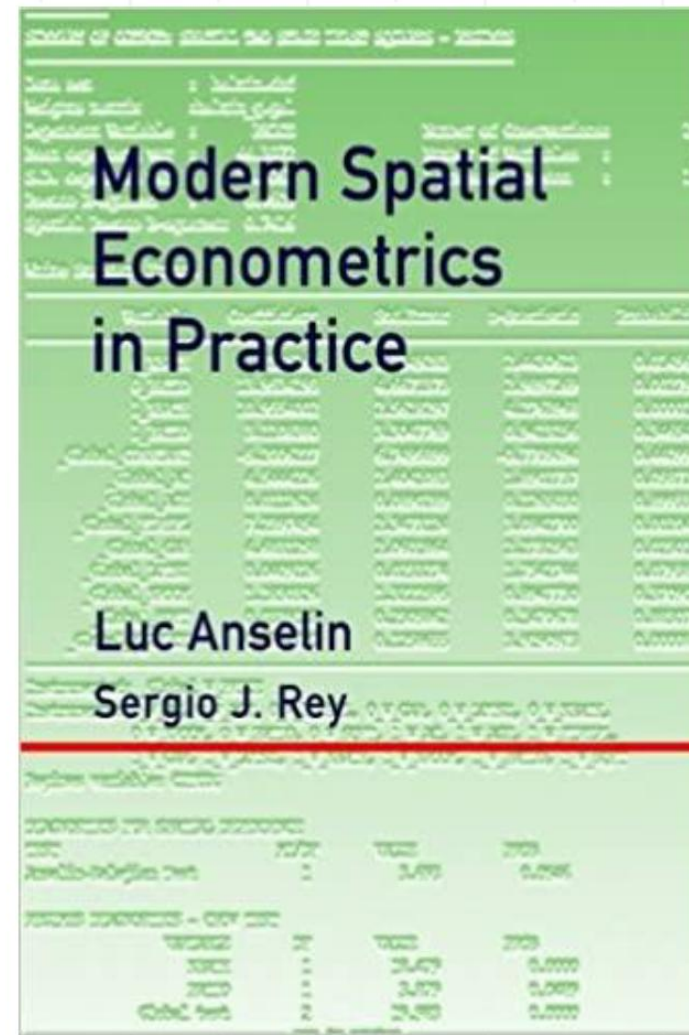


spatial lag model

$$Y = \rho WY + X\beta + \varepsilon$$

- Spatial lag models are typically used when we see spatial autocorrelation of the dependent variable Y .
- This means that neighboring outcomes Y_j will have an impact on Y_i itself.
- We're again violating an assumption in standard regression models (exogeneity) – our estimators are biased...
- But pysal has some remedies for this.
- Use When: You suspect or detect that the dependent variable in one location is influenced by the values of the dependent variable in neighboring locations.





PySAL and Other Options

- An alternative to PySAL for the spatial regression analysis is GeoDaSpace
 - <https://spatial.uchicago.edu/software>
- For further instruction on these topics, check the open course on Spatial Regression by Luc Anselin
 - <https://youtu.be/50kc3F0aE6Y>



Free Online Book

- *Geographic Data Science with PySAL and the PyData Stack*
 - by Sergio J. Rey, Dani Arribas-Bel and Levi J. Wolf
 - Site for the book <https://geographicdata.science/book/intro.html>
- Introduction to Geospatial Data Analysis with Python | SciPy 2018 Tutorial
<https://youtu.be/kJXUUO5M4ok>



Linear Regression Assumptions (cont.)

- **Exogeneity ($E[u|X] = 0$):** The error term should not be correlated with the independent variables; otherwise, leads to **biased and inconsistent** OLS estimates

Use Instrumental Variables (IV) Regression

Use Fixed Effects Models (for Panel Data)

Control for Omitted Variables

Other quasi-experimental design methods (DIDs, dynamic DIDs, etc)